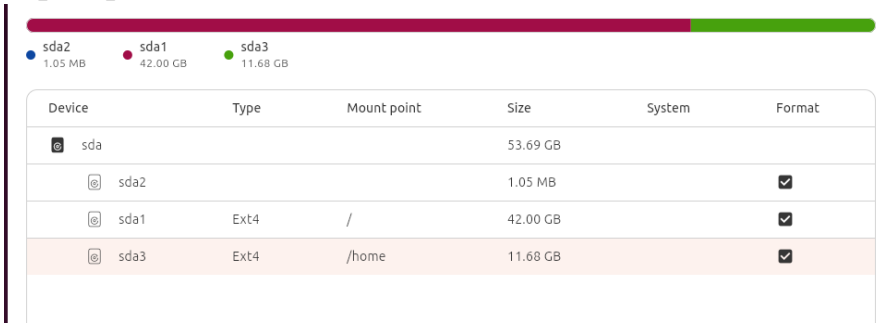
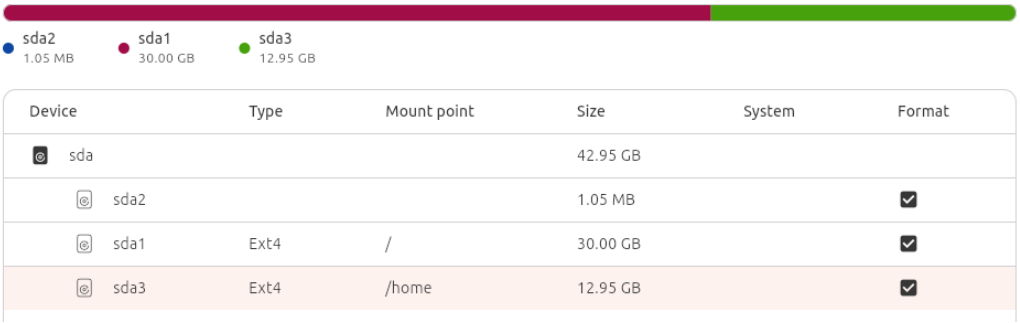
Работа с VM

1. Одна из VM1 (называется devops) у меня уже была создана, при выполнении ДЗ «Введение в DevOps2.Операционные системы. Часть 1». Прикладываю повторно скрин разметки диска:



2. Не буду прикладывать скрины полной установки, а только разметку диска. Итак, создаю вторую VM2 с названием Ubuntu_Dev. Прикладываю скриншот разметки диска:



3. Также ранее создавала ssh соединение между хостовой ОС и VM1 (devops), есть даже мини скрипт для подключения, чтобы не вводить постоянно:

```
anastasia@DESKTOP-ADT12D6:~/scripts$ bat ssh-ubuntu.sh
File: ssh-ubuntu.sh
1  ssh -p 22 anastasia@192.168.203.131
```

4. После запуска скрипта ввожу пароль от VM1 (devops) и успешно подключаюсь:

```
anastasia@DESKTOP-ADT12D6:~/scripts$ ./ssh-ubuntu.sh
anastasia@192.168.203.131's password:
Welcome to Ubuntu 25.10 (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

20 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Last login: Wed Apr  8 18:31:09 2026 from 192.168.203.1
anastasia@devops:~$
```

5. Покажу как пройти все этапы для подключения на примере VM1 (devops) -> Хостовая ОС. Для этого сначала на хостовой ОС ввожу команду ip a для вывода айпи, который необходим будет для подключения:

```
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group default qlen 1000
    link/ether 00:15:5d:72:bb:8a brd ff:ff:ff:ff:ff:ff
    inet 172.29.190.204/20 brd 172.29.191.255 scope global eth0
       valid_lft forever preferred_lft forever
```

6. Перехожу в VM1 (devops), далее необходимо создать файл, куда будет записывать ключ, для этого я создаю файл id_rsa в созданную папку ssh-keys/

```
anastasia@devops:~$ eza -la ssh-keys/
.rw-rw-r-- 0 anastasia 17 Apr 13:55 id_rsa
```

7. Теперь генерирую ключ в созданный в предыдущем пункте с помощью команды sudo ssh-keygen –t rsa (метод шифрования) –b 4096 (биты). Соглашаюсь и ввожу passphrase, после чего выводит, что сгенерировался ключ:

```
anastasia@devops:~$ sudo ssh-keygen -t rsa -b 4096
Generating public/private rsa key pair.
Enter file in which to save the key (/root/.ssh/id_rsa): ssh-keys/id_rsa
ssh-keys/id_rsa already exists.
Overwrite (y/n)? y
Enter passphrase for "ssh-keys/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in ssh-keys/id_rsa
Your public key has been saved in ssh-keys/id_rsa.pub
The key fingerprint is:
SHA256:uGFTnbGBQRdlyNxcI+hLkSGFSvoGphYCbNaSxIc88zE root@devops
The key's randomart image is:
+---[RSA 4096]----+
|=.+     .+BBX+.o  |
|.% E   . o+B*o. .  |
|+ * oo ...+.       |
|. ..+ .o  o        |
| . + o= S. .       |
|  o  .o+  .        |
| .   ..            |
|                   |
|                   |
+----[SHA256]-----+
```

```
anastasia@devops:~$ eza -l ssh-keys/
.rw-rw-r-- 3.4k anastasia 17 Apr 14:01 id_rsa
.rw-r--r--  737 root      17 Apr 14:01 id_rsa.pub
```

8. Теперь нужно передать (скопировать) открытый ключ на хостовую ОС, для этого ввожу команду ssh-copy-id –p 22 anastasia@172.29.190.204. Затем подключаюсь для проверки к хостовой ОС с помощью команды ssh –p 22 anastasia@172.29.190.204, затем ввожу пароль и вот результат:

```
anastasia@devops:~/scripts$ ssh -p 22 anastasia@172.29.190.204
anastasia@172.29.190.204's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Apr 17 11:13:08 MSK 2026

  System load:  0.01                Processes:             50
  Usage of /:   0.6% of 1006.85GB   Users logged in:       1
  Memory usage: 3%                  IPv4 address for eth0: 172.29.190.204
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Fri Apr 17 11:13:09 2026 from 172.29.176.1
anastasia@DESKTOP-ADT12D6:~$
```

9. Перехожу к созданию соединения между VM2 (Ubuntu_Dev) -> VM1 (devops). Чтобы не проделывать одни и те же действия для каждого соединения, я написала скрипт на bash. Полный итог на скриншоте:

```
nastya@nastya-vm:~/scripts$ ./ssh-VM1
touch: cannot touch 'ssh-keys/id_rsa': No such file or directory
Generating public/private rsa key pair.
Your identification has been saved in /home/nastya/ssh-keys/id_rsa
Your public key has been saved in /home/nastya/ssh-keys/id_rsa.pub
The key fingerprint is:
SHA256:y0kTD3cdWFAIrCXcbhuukKBU1jYy/K5oE2oYLzUDRH8 root@nastya-vm
The key's randomart image is:
+---[RSA 4096]----+
|.. . . . o...*+  |
| .. * + o + o. . |
|.  + E .o=. . .  |
|. . o . .=+.     |
| o . o .So.o     |
|. *   +o +o      |
|.= = . .+.       |
|+.= .   .        |
|.o .             |
+----[SHA256]-----+
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/nastya/ssh-keys/id_rsa.pub"
The authenticity of host '192.168.203.131 (192.168.203.131)' can't be established.
ED25519 key fingerprint is SHA256:uGADtUF3mvT2XPavUJKkgOJec/xqk1Prz2kePAqiz0g.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
anastasia@192.168.203.131's password:

Number of key(s) added: 1

Now try logging into the machine, with: "ssh -i /home/nastya/ssh-keys/id_rsa -p 22 'anastasia@192.168.203.131'"
and check to make sure that only the key(s) you wanted were added.

nastya@nastya-vm:~/scripts$ ssh -p 22 anastasia@192.168.203.131
The authenticity of host '192.168.203.131 (192.168.203.131)' can't be established.
ED25519 key fingerprint is SHA256:uGADtUF3mvT2XPavUJKkgOJec/xqk1Prz2kePAqiz0g.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.203.131' (ED25519) to the list of known hosts.
anastasia@192.168.203.131's password:
Welcome to Ubuntu 25.10 (GNU/Linux 6.17.0-22-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

20 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Last login: Fri Apr 17 13:46:44 2026 from 192.168.203.1
anastasia@devops:~$
```

10. И последнее соединение VM2 (Ubuntu-Dev)-> Хостовая ОС. Также через скрипт

```
The key fingerprint is:
SHA256:oQW2APKtzB18OYWNrQ7O7f6dHiPlkEJWDLGe19YPXZc root@nastya-vm
The key's randomart image is:
+---[RSA 4096]----+
|. ... =Oo        |
| o o ooB+       .|
|  . + O.o      Eo|
| o o.*.= + . . ..|
|  +o.+= S + o .  |
|    o oo =   o   |
|     .  . +   .  |
|      .  o +     |
|     .....+      |
+----[SHA256]-----+
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/nastya/ssh-keys/id_rsa_host.pub"
The authenticity of host '172.29.190.204 (172.29.190.204)' can't be established.
ED25519 key fingerprint is SHA256:sa4RpJqa6dcajSzCsn5UNjHDr/J2tGbUxlYTRM8vr4M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
anastasia@172.29.190.204's password:

CDROM y(s) added: 1

Now try logging into the machine, with: "ssh -i /home/nastya/ssh-keys/id_rsa_host -p 22 'anastasia@172.29.190.204'"
and check to make sure that only the key(s) you wanted were added.

nastya@nastya-vm:~/scripts$ ssh -p 22 anastasia@172.29.190.204
The authenticity of host '172.29.190.204 (172.29.190.204)' can't be established.
ED25519 key fingerprint is SHA256:sa4RpJqa6dcajSzCsn5UNjHDr/J2tGbUxlYTRM8vr4M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.29.190.204' (ED25519) to the list of known hosts.
anastasia@172.29.190.204's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Fri Apr 17 14:27:40 MSK 2026

  System load:  0.02               Processes:             49
  Usage of /:   0.7% of 1006.85GB  Users logged in:       1
  Memory usage: 3%                 IPv4 address for eth0: 172.29.190.204
  Swap usage:   0%

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge
Last login: Fri Apr 17 11:13:33 2026 from 172.29.176.1
anastasia@DESKTOP-ADT12D6:~$
```

11. Эксперимент с iptables, например я хочу, чтобы к хостовой ОС не могла подключиться ни одна ВМ по ssh по порту 22. Делаю с помощью команды sudo iptables –A INPUT –p tcp –dport 22 –j DROP. Затем проверяем применилось ли правило. Проверить что правило применилось можно вывести список sudo iptables -L:

```
anastasia@DESKTOP-ADT12D6:~$ sudo iptables -L
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
DROP       tcp  --  anywhere             anywhere             tcp dpt:ssh
```

12. Проверяем подключение с VM2 (Ubuntu-Dev) на хостовую ОС. Итог:

```
nastya@nastya-vm:~/scripts$ ssh -p 22 anastasia@172.29.190.204
ssh: connect to host 172.29.190.204 port 22: Connection refused
nastya@nastya-vm:~/scripts$
```